IOT BASED
MORSE CODE SIGNALLING

LEGENDARY TITANS

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Abstract

- This project is a simple Morse code signaling system using Arduino, an LED, and a buzzer.
- It changes messages like "SOS" into light and sound signals using Morse code.
- The system can be turned on remotely using Wi-Fi or Bluetooth.
- It is useful in emergencies, for quiet alerts, or in smart homes.
- The project is low-cost, easy to build, and helpful when normal communication (like phones) is not working

Objective

- Develop an IoT-based Morse code transmission system for efficient communication.
- Implement a real-time Morse code translator for IoT devices.
- Enhance emergency communication systems using Morse code and IoT.
- Achieve reliable Morse code transmission over long distances using IoT.
- Integrate Morse code signalling with IoT sensors for smart applications.

Project Overview

- The IoT-Based Morse Code Signaling System is a communication device that utilizes Morse code and Internet of Things (IoT) technology to enable real-time message transmission over the internet.

- The system is designed to take user input typed characters—convert them into Morse code, and transmit the encoded message to a remote device or platform via LED.

- This project is about creating a Morse code alert system using Ardiuno, an LED and a Buzzer.

Solution Overview

- The system consists of an input module to enter Morse code, a microcontroller with Wi-Fi to process and transmit the data, and a cloud platform (like Arduino IDE) to send messages to a remote receiver.
- At the receiving end, the Morse code is decoded and presented using visual or audio signals (LEDs, buzzers) or displayed as readable text on a screen or mobile app.
- This system can be operated by C++ programming language.
- The solution bridges traditional Morse signaling with modern IoT infrastructure, ensuring long-range, reliable communication without relying on voice or complex hardware.

Applications

- ✓ Emergency Communication in Remote Areas.
- ✓ Enabling communication for people with disabilities
- ✓ Assistive Technology for Speech-Impaired Individuals.
- ✓ Military and Defense.
- ✓ Disaster Response
- ✓ Wearable Technology.

Components Used

- Arduino UNO
- LED
- Buzzer
- Resistor
- Breadboard
- Jumper wire

Architecture Diagram

[User Input Device]

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[Mobile App/Web Interface] ⟷ [Wi-Fi Module (ESP8266/ESP32)]

[Cloud/MQTT Broker]

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[Wi-Fi Module (ESP8266/ESP32)]

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[Arduino Microcontroller]

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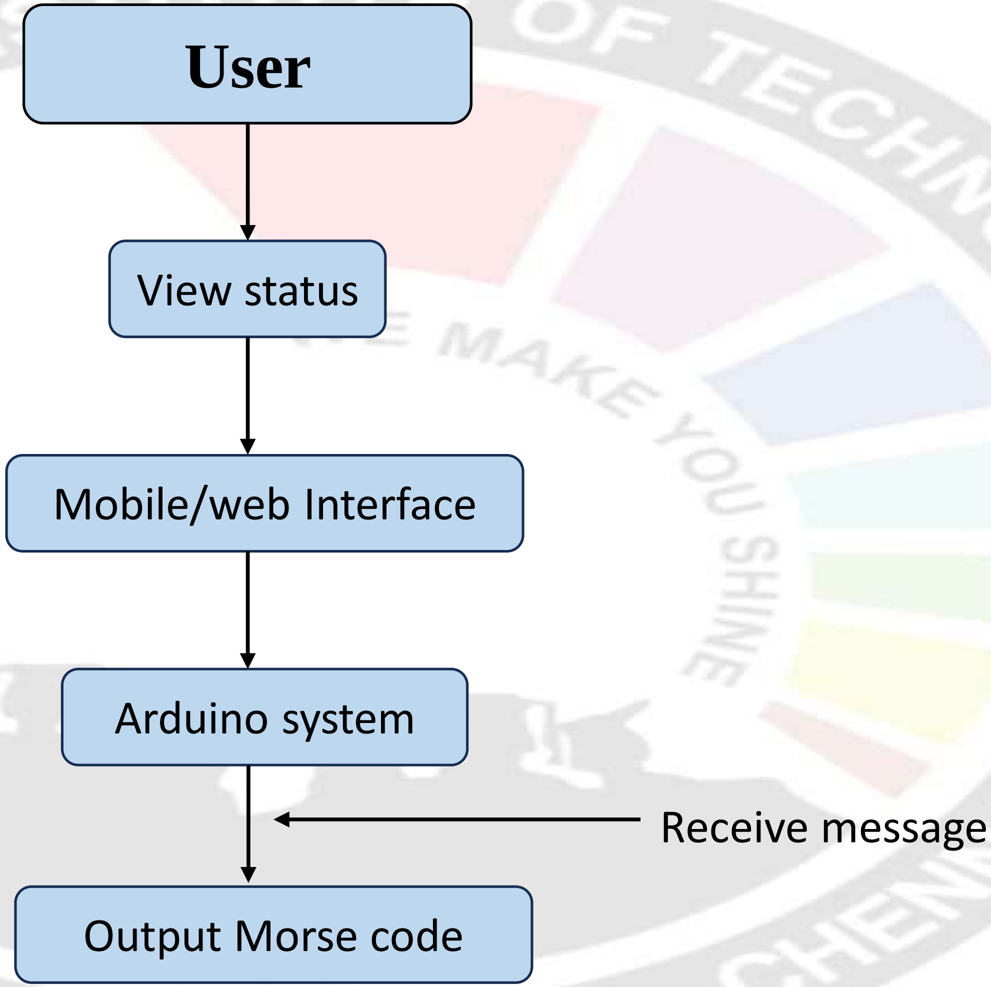
[Morse Code Output Device(s)]

(LED / Buzzer / Servo Motor for signaling)

Case diagram



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THANK YOU

Reference:
Youtube